Blueprint

The Clock That Never Ticks: a machine-checked path from the dilogarithm to quantum speed limits in Lean 4

Special functions · the dilogarithm and the Clausen function

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Statements

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At a glance

What the paper contributes

A sorry-free Lean 4 path from the dilogarithm, which Mathlib cites but does not define, to a quantum state that never orthogonalizes.

- **The real dilogarithm** with Euler's reflection, Landen's transformation, the duplication formula and $\mathrm{Li}_2(1/2)$ (Theorem 1); the golden-ratio values and the Lee–Yang $c_{\mathrm{eff}} = 2/5$ from a 3×3 linear system, without the five-term relation.
- **The circle**: the Clausen function and Catalan's constant $G = \mathrm{Cl}_2(\pi/2)$; the Fejér–Jackson inequality (Theorem 2) and, by Abel summation without asymptotics, $\mathrm{Cl}_2(\theta) \geq \sin(\theta)/2$ on $(0, \pi)$.
- **Speed limits**: Margolus–Levitin and an L^1 form of Mandelstam–Tamm for a scalar spectral measure (Theorem 3).
- **The assembly** (Theorem 4): populations $6/(\pi^2 n^2)$ on levels $E_n = n$ give infinite mean energy and variance, so both bounds are silent, and yet the state never reaches an orthogonal state.
- The mathematics is classical; the paper claims the formalization and the assembly, and records the Eneström–Kakeya reading as an observation (Remark 1).

🚀 **Start here: Theorem 1**: Euler reflection, Landen, duplication and $\mathrm{Li}_2(1/2)$ for the real dilogarithm.

Main results:

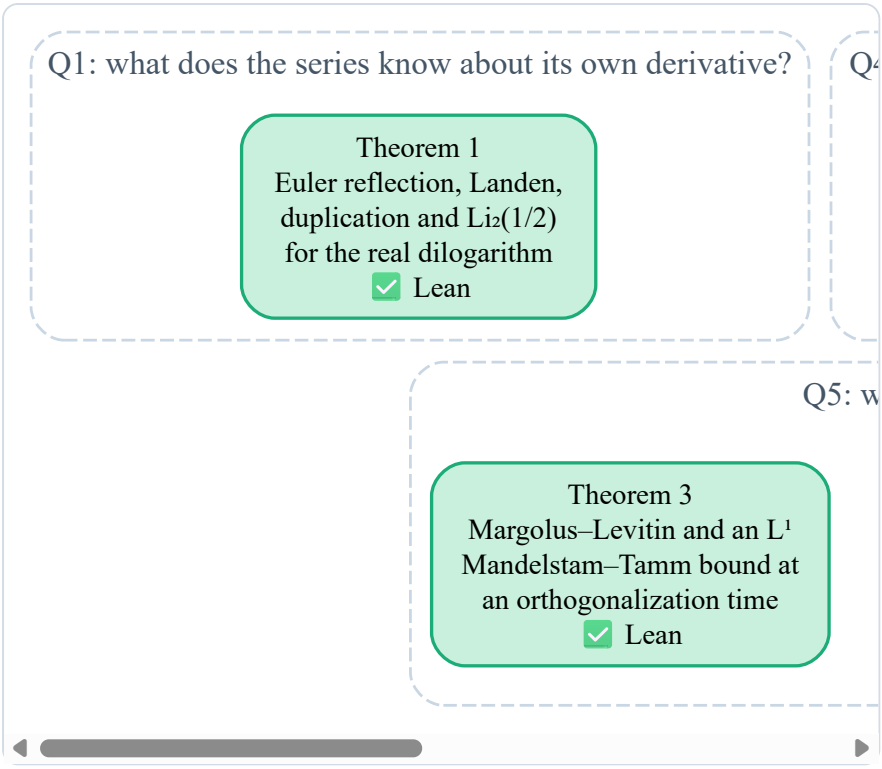
- **Theorem 4**: the weight-2 zeta state never reaches an orthogonal state

5 results | ✅ **4** entirely in Lean | ◐ **1** partly in Lean

💻 **0** checked by a program | written proof only (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 **Setting (copied from the paper, §1)**

The file `Mathlib/Analysis/SpecialFunctions/Integrals/LogTrigonometric.lean` of Mathlib … describes one of its integrals by referring to the dilogarithm

$$\mathrm{Li}_2(x) \;=\; \sum_{n\ge 1} \frac{x^n}{n^2}.$$

(§3) In the Rogers normalization $\mathrm{L}(x) = \mathrm{Li}_2(x) + \frac{1}{2}\ln x \ln(1-x)$ the $\ln^2$ defects cancel.
(§4) On the circle the dilogarithm splits into a tame half and a wild half:

$$\mathrm{Li}_2(e^{i\theta}) \;=\; \underbrace{\sum_{n\ge 1} \frac{\cos n\theta}{n^2}}_{=\,\frac{\pi^2}{6} - \frac{\pi\theta}{2} + \frac{\theta^2}{4}} \;+\; i \underbrace{\sum_{n\ge 1} \frac{\sin n\theta}{n^2}}_{=\,\mathrm{Cl}_2(\theta)} \qquad (\theta \in [0, 2\pi]).$$

(§6) A state $|\psi_0\rangle = \sum_n c_n |E_n\rangle$ evolves with autocorrelation $S(t) = \sum_n p_n e^{-iE_n t}$, $p_n = |c_n|^2$, and reaches an orthogonal state at time τ iff $S(\tau) = 0$. Take populations $p_n = \frac{6}{\pi^2 n^2}$ on levels $E_n = n$: the *weight-2 zeta state*.

2 Q1: what does the series know about its own derivative?

Theorem 1. Lean (`Li₂_add_Li₂_one_sub`, `Li₂_landen`, `Li₂_sq`, `Li₂_one_half`) ✓ p. 3

> *On the stated domains,*

$$\begin{aligned}
\mathrm{Li}_2(x)+\mathrm{Li}_2(1-x) &= \tfrac{\pi^2}{6}-\ln x\,\ln(1-x) && x\in(0,1)\\
\mathrm{Li}_2(x)+\mathrm{Li}_2\!\left(\tfrac{x}{x-1}\right) &= -\tfrac{1}{2}\ln^2(1-x) && x\in(0,\tfrac{1}{2})\\
\mathrm{Li}_2(x^2) &= 2\big(\mathrm{Li}_2(x)+\mathrm{Li}_2(-x)\big) && |x|\le 1\\
\mathrm{Li}_2(\tfrac{1}{2}) &= \tfrac{\pi^2}{12}-\tfrac{\ln^2 2}{2}.
\end{aligned}$$

Lean · ours show

- `Dilog.Li₂_add_Li₂_one_sub`
- `Dilog.Li₂_landen`
- `Dilog.Li₂_sq`
- `Dilog.Li₂_one_half`
- `Dilog.Li₂_inv_goldenRatio_sq`
- `Dilog.Li₂_inv_goldenRatio`
- `Dilog.Li₂_neg_inv_goldenRatio`
- `Dilog.rogersL_inv_goldenRatio_sq`
- `Dilog.rogersL_gold_sum`

✅ Checked when this page was built: Basic.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — Li₂_add_Li₂_one_sub, Li₂_landen, Li₂_sq, Li₂_one_half, Li₂_inv_goldenRatio_sq, Li₂_inv_goldenRatio, Li₂_neg_inv_goldenRatio, rogersL_inv_goldenRatio_sq, rogersL_gold_sum: propext, Classical.choice, Quot.sound.
Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean Basic.lean` (SHA-256 of the compiled file: `ca33594f06b52c38…`; a source page on this site may show it with comments edited, under its own hash).

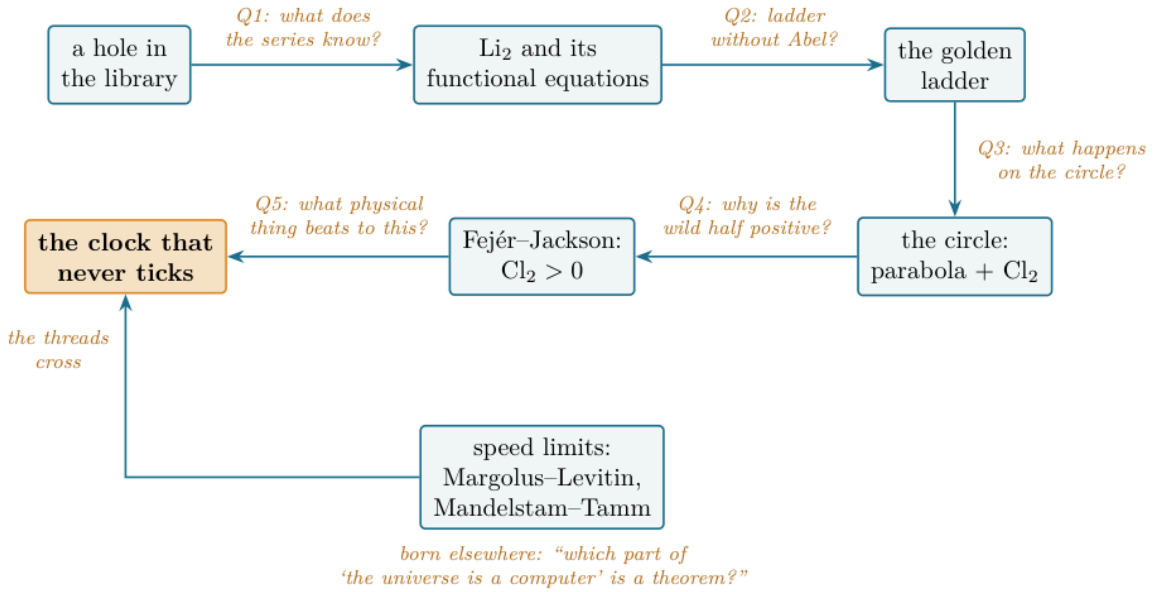
🆕 **Classical and new:** classical (Euler 1768, Landen 1780; Lewin, Zagier); the paper claims, from its June 2026 checks, the first formalization of the real dilogarithm and these identities in a proof assistant (§7(i)).

↳ Not used by another result of the paper.

Proof

Formalized in Lean (see the linked declarations).

Figure 1



The honest map. The dilogarithm thread runs along the top and back; the quantum thread (bottom) was born independently and *crossed* it at the end — the crossing, not the chain, is where the title theorem lives. Every node is a block of `sorry`-free Lean.

5 Q4: why is the wild half positive?

Theorem 2. Lean (`fejerSum_pos`)✓ p. 5

> *For every* $M \geq 1$ *and* $\theta \in (0, \pi)$: $\displaystyle\sum_{k=1}^{M} \frac{\sin k\theta}{k} > 0.$

Lean · ours show

- `Dilog.fejerSum_pos`
- `Dilog.Cl₂_pos`
- `Dilog.Cl₂_pi_div_two`
- `Dilog.hasSum_cos_div_sq`

✅ Checked when this page was built: Clausen.lean, FejerJackson.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — fejerSum_pos, Cl₂_pos, Cl₂_pi_div_two, hasSum_cos_div_sq: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean Clausen.lean FejerJackson.lean` (SHA-256 of the compiled file: `78931a748edb7c0c`…; a source page on this site may show it with comments edited, under its own hash).

NEW **Classical and new:** classical (Fejér 1910, Jackson 1911); the paper records two proof-engineering choices, the division-free Dirichlet telescope and the asymptotics-free bound $\mathrm{Cl}_2(\theta) \geq \sin(\theta)/2$ (§7(ii)).

↳ Used in: Theorem 4 (p. 7)

Proof

> Formalized in Lean (see the linked declarations).

6 Q5: what physical thing beats to this function?

Theorem 3. Lean (`margolus_levitin`, `mandelstam_tamm_L1`; $\hbar = 1$)✓ p. 7

> *If* $S(\tau) = 0$ *with* $\tau \geq 0$, *then*
>
> $$\pi \;\leq\; 2\,\langle E\rangle\,\tau \qquad \text{and} \qquad 1 \;\leq\; D_1\,\tau, \quad D_1 = \textstyle\sum_n p_n |E_n - \bar{E}|.$$

Lean · ours show

- `QSL.margolus_levitin`
- `QSL.mandelstam_tamm_L1`

✅ Checked when this page was built: MandelstamTamm.lean, MargolusLevitin.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — margolus_levitin, mandelstam_tamm_L1: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean MandelstamTamm.lean MargolusLevitin.lean` (SHA-256 of the compiled file: `224c15dcafc7d45b…`; a source page on this site may show it with comments edited, under its own hash).

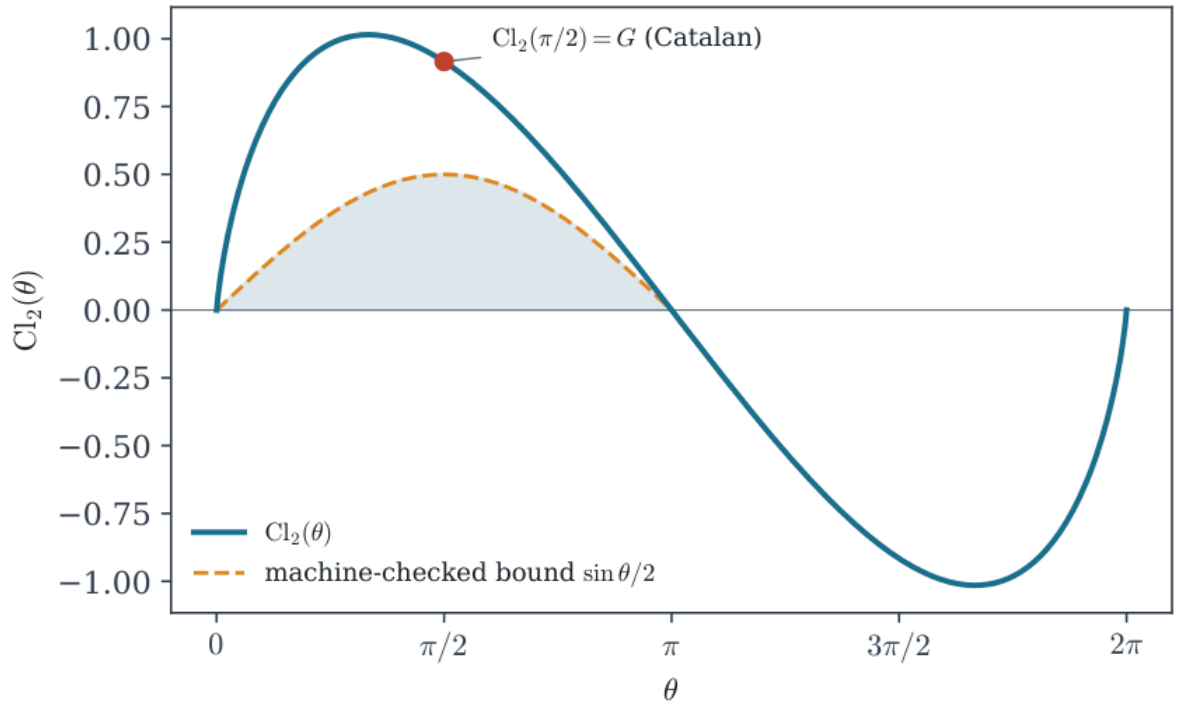
🆕 **Classical and new:** classical (Mandelstam–Tamm 1945, Margolus–Levitin 1998); the paper states them for a scalar spectral measure and calls the fully real statements a reusable choice (§7(ii)). The sharp π/(2ΔE) is left open (§8).

↳ Not used by another result of the paper.

Proof

Formalized in Lean (see the linked declarations).

Figure 2



The Clausen function on $[0, 2\pi]$ with the machine-checked lower bound $\sin(\theta)/2$ (shaded) on $(0, \pi)$, and Catalan's constant at the quarter circle. The generating script asserts the bound on a 400-point grid at 30 digits.

Theorem 4. Lean (`zetaState_never_orthogonal`)✓ p. 7

> *For every $\theta \in (0, 2\pi)$, the real and imaginary parts of the weight-2 zeta-state autocorrelation are not both zero: the state never reaches an orthogonal state.*

Lean · ours **show**

- `Dilog.zetaState_never_orthogonal`
- `Dilog.Cl₂_two_pi_sub`

✅ Checked when this page was built: Clausen.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — zetaState_never_orthogonal, Cl₂_two_pi_sub: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean Clausen.lean` (SHA-256 of the compiled file: `6d0145e090a18399`…; a source page on this site may show it with comments edited, under its own hash).

NEW **Classical and new:** the assembly is the paper's: «whose physical reading we have not seen stated, though we would not be surprised to find it folklore among polylogarithm or QSL specialists» (§7(iii)).

↳ Used in: 1 (p. 7)

Proof

> Formalized in Lean (see the linked declarations).

Remark 1. (Which clocks do tick: a closing observation) p. 8

For monotone non-increasing populations on equally spaced levels, ticking (orthogonalization) means a zero of the power series $\sum_n p_n z^n$ *on* the unit circle. The Eneström–Kakeya theorem (1893/1912) [12] already forbids zeros inside the disc; its boundary case pins down the rest. From $(1-z)\sum_k p_k z^k = p_0 + \sum_{k\geq 1}(p_k - p_{k-1})z^k - p_N z^{N+1}$, the triangle inequality at $|z| = 1$ is an equality only if every nonzero decrement $p_{k-1} - p_k$ sits at an index k with $z^k = 1$: the decrements must concentrate on an arithmetic progression, and the zero is a root of unity. The Margolus–Levitin–sharp state $(|0\rangle + |E\rangle)/\sqrt{2}$ is exactly such a case ($\frac{1}{2}(1+z)$, zero at $z=-1$). The zeta state is the opposite extreme: its successive decrements $p_n - p_{n+1} > 0$ at *every* index, never confined to a progression, so no root of unity survives and the clock cannot tick — recovering Theorem 4 from the coefficient side. We have not found the resulting dictionary — root location of positive-coefficient series ↔ orthogonalizability of monotone quantum states — in either literature; we record it as an observation and a road.

Lean · ours, in part **show**

added to the Lean repository after publication (21-oct-2026): the Eneström–Kakeya boundary case behind the remark: positive non-increasing coefficients give no zero in the open disk, zeros on the circle are roots of unity, none with strictly decreasing coefficients; and monotone states never become orthogonal

- `Dilog.enestrom_kakeya`
- `Dilog.enestrom_kakeya_boundary`
- `Dilog.enestrom_kakeya_strict`
- `Dilog.monotoneState_never_orthogonal`

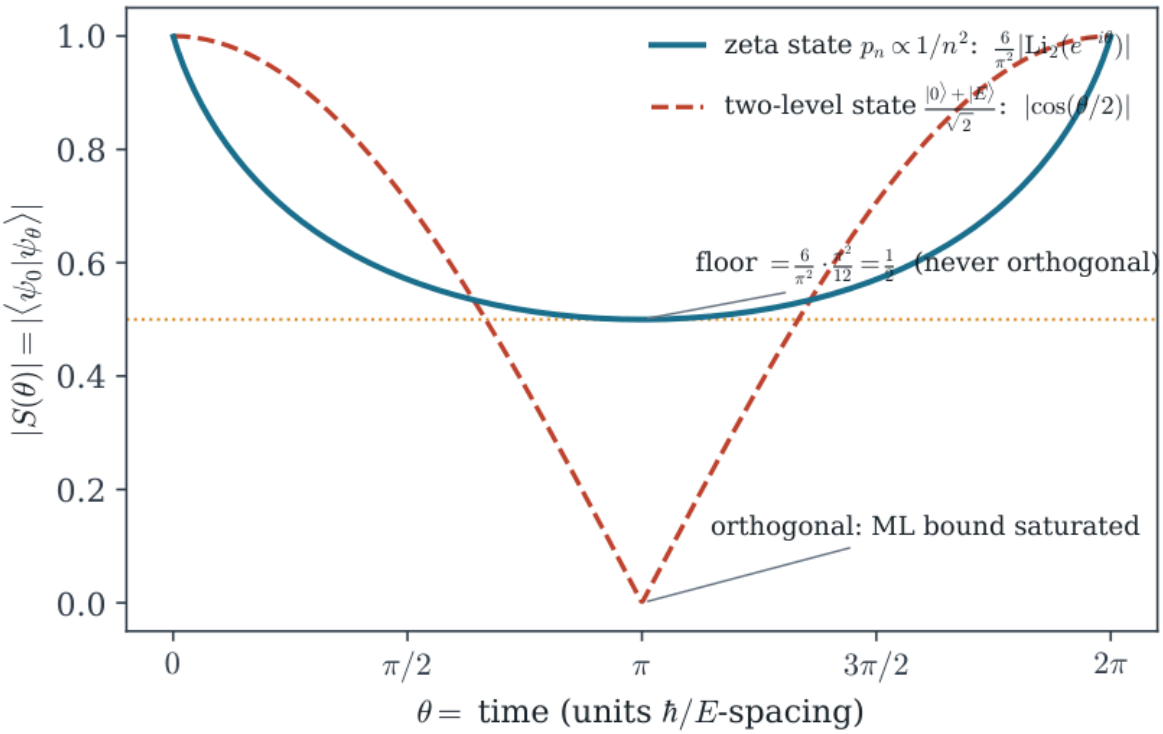
✅ Checked when this page was built: EnestromKakeya.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — enestrom_kakeya, enestrom_kakeya_boundary, enestrom_kakeya_strict, monotoneState_never_orthogonal: propext, Classical.choice, Quot.sound.

Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean EnestromKakeya.lean` (SHA-256 of the compiled file: `53f569d625086f34`…; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** the paper has «not found the resulting dictionary … in either literature» and records it «as an observation and a road» (Remark 1); it is not formalized.

↳ Not used by another result of the paper.

Figure 3



Two states with the same level spacing. The two-level state $(|0\rangle + |E\rangle)/\sqrt{2}$ (dashed) saturates Margolus–Levitin and strikes orthogonality at $\theta = \pi$. The zeta state (solid), with *infinite* mean energy, has $|S| \geq \frac{1}{2}$: the clock that never ticks. The script asserts the floor equals $\eta(2) = \pi^2/12$ at $\theta = \pi$.

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